

DSA-1 Lab Practice Sheet: Searching & Recursion

Searching Problems

1. Linear Search: Count Numbers Greater Than Given Value

Given an array of integers and a number X , count how many elements in the array are greater than X .

Example:

Input: Array = [2, 5, 1, 8, 3], $X = 3$

Output: 2

Explanation: Numbers greater than 3 are 5 and 8.

2. Binary Search in an Ascending Array

Given a sorted array in ascending order and a target value, find the index of the target using binary search.

Example:

Input: Array = [1, 3, 5, 7, 9], Target = 5

Output: 2

3. Binary Search in a Descending Array

Given a sorted array in descending order and a target value, find the index using binary search.

Example:

Input: Array = [9, 7, 5, 3, 1], Target = 5

Output: 2

4. Find First Occurrence of a Number

Given a sorted array (may contain duplicates), find the first occurrence of a given number.

Example:

Input: Array = [1, 2, 2, 2, 3, 4], Target = 2

Output: 1

5. Find Last Occurrence of a Number

Given a sorted array (may contain duplicates), find the last occurrence of a given number.

Example:

Input: Array = [1, 2, 2, 2, 3, 4], Target = 2

Output: 3

6. Find Insert Position

Given a sorted array and a target value, return the index where the target is found. If not found, return the index where it should be inserted.

Example:

Input: Array = [1, 3, 5, 7], Target = 4

Output: 2

Recursion Problems

1. Summation from 1 to n

Find the sum of all numbers from 1 to n using recursion.

Example:

Input: $n = 5$

Output: 15

2. Factorial of n

Find the factorial of a number n using recursion.

Example:

Input: $n = 4$

Output: 24

3. Print Elements of an Array

Given an array, print all elements using recursion.

Example:

Input: [1, 2, 3]

Output: 1 2 3

4. Print Elements in Reverse

Given an array, print all elements in reverse order using recursion.

Example:

Input: [1, 2, 3]

Output: 3 2 1

5. Search an Element

Given an array and a target value, check if the element exists using recursion.

Example:

Input: Array = [4, 7, 1, 9], Target = 7

Output: Found

6. Sum of Digits

Given a number, find the sum of its digits using recursion.

Example:

Input: 1234

Output: 10